



Forensic audio analysis

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*Dr. Edmond Locard Biography And
Legacy*

Forensic Science

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Forensic science is the use of scientific methods or expertise to investigate crimes or examine evidence that might be presented in a court of law. Forensic science comprises a diverse array of disciplines, from fingerprint and DNA analysis to anthropology and wildlife forensics.

History of Forensic Science



Although it is uncertain exactly where the concept of forensic science originated, most historical experts agree it was very likely in China around the 6th century or earlier. This belief is based on the earliest known mention of the concept, found in a book titled “Ming Yuen Shih Lu,” printed in that period.

Father of modern forensic Science



Dr. Edmond Locard (13 December 1877 – 4 May 1966)[1] was a French criminologist, the pioneer in forensic science who became known as the "Sherlock Holmes of France". He formulated the basic principle of forensic science: "Every contact leaves a trace". This became known as Locard's exchange principle.

Biography

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Locard was born in Saint-Chamond, France on December 13, 1877, although some records claim he was born in 1872. He studied medicine and law at Lyon, France, eventually becoming the assistant of Alexandre Lacassagne, a criminologist and professor. He held this post until 1910, when he began the foundation of his criminal laboratory. His lab, located in Lyon, was the first forensic lab in Europe.

In 1910, Locard succeeded in persuading the Police Department of Lyon to give him two attic rooms and two assistants, to start what became the first police forensic laboratory.

Locard's daughter Denise would be born on November 18, 1917 in Paris.

*Locard produced a monumental, seven-volume work, *Traité de Criminalistique*. He also was first to codify Galton points, fingerprint characteristics meant for identification.*

Locard continued his research in Lyon until his death in 1966

Legacy

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The young Georges Simenon, later to become a well-known detective writer, is known to have attended some Locard lectures in 1919 or 1920.

Locard is considered to be the father of modern forensic science. His Exchange Principle is the basis of all forensic work; the principle stipulates that when any two objects come into contact, there is always a transference of material between each object.

In November 2012, he was nominated to the French Forensic Science Hall of Fame of the Association Québécoise de Criminalistique.

Forensic Audio Analysis

History of Audio Forensics

Audio Forensics

Types of Audio Forensic Investigations

Spectrogram

Forensic Investigations:

Audio Analysis Expert

Analysis of Audio Evidence

Forensic Speech Analysis Expertise

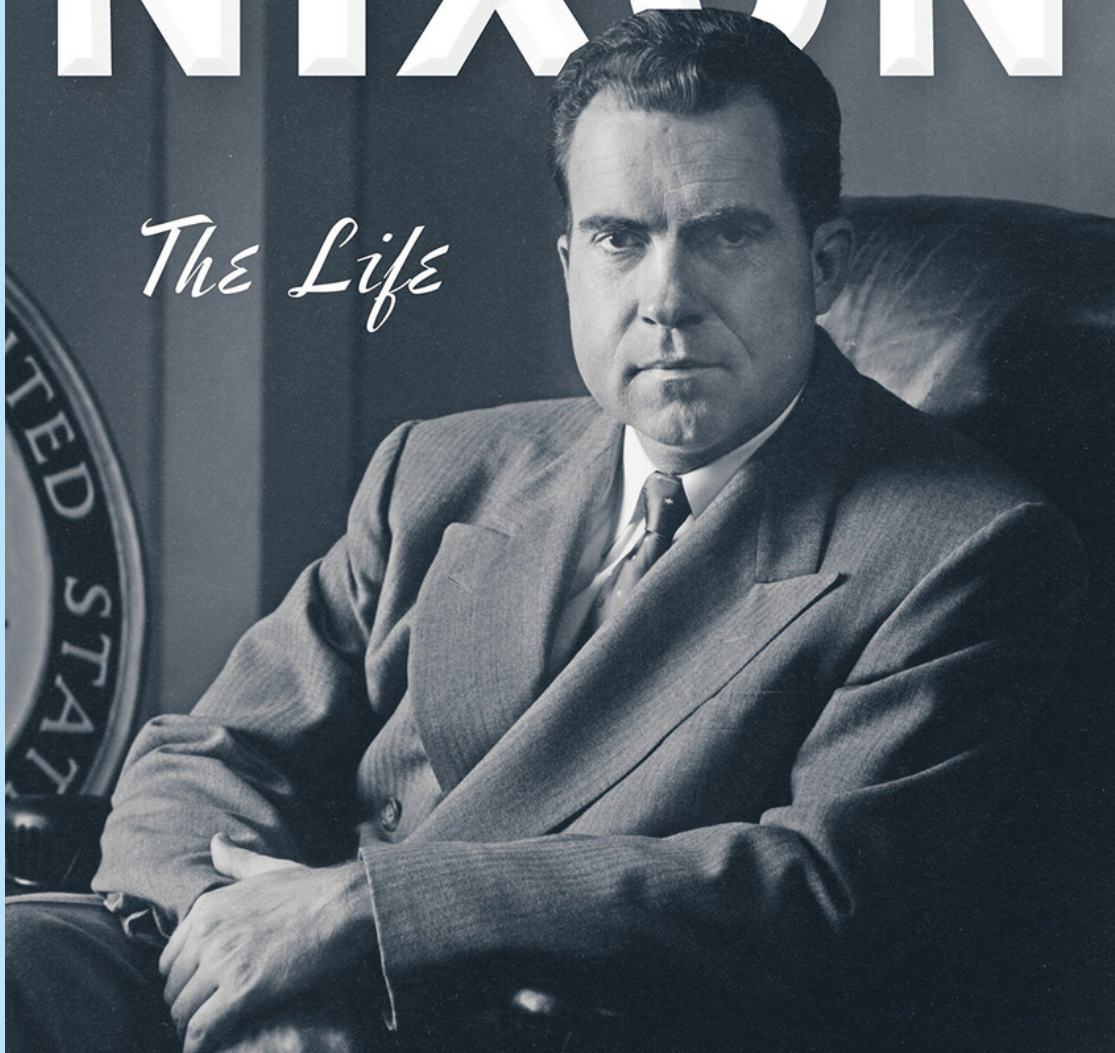
History of Audio Forensics



The field of audio forensics was largely established in 1973 during the Watergate scandal. A federal court commissioned a panel of audio engineers to investigate the gaps in President Nixon's Watergate Tapes, which were secret recordings U.S. President Richard Nixon made while in office. The probe found nine separate sections of a vital tape had been erased. The report gave rise to new techniques to analyze magnetic tape

RICHARD NIXON

The Life



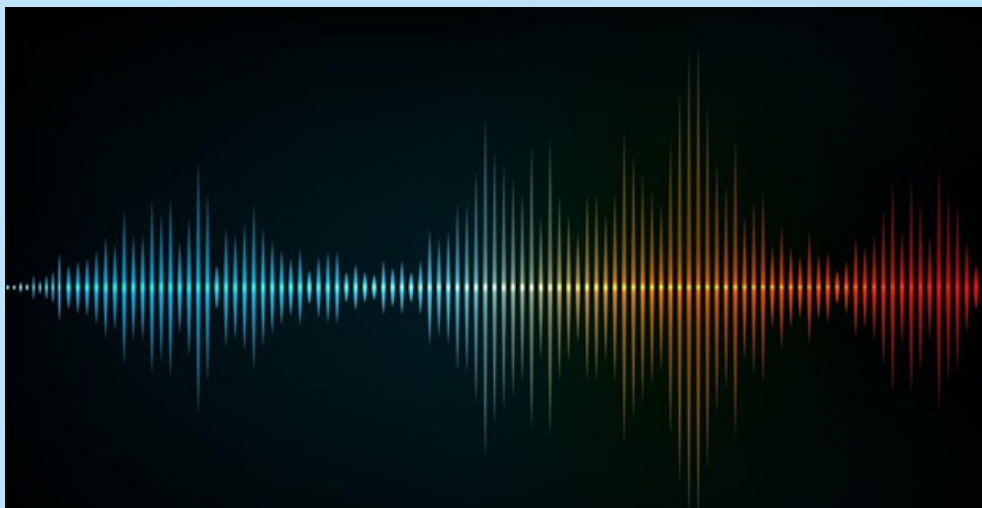
U.S. President Richard Nixon

Audio Forensics

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Audio Forensics is the field of forensic science relating to the acquisition, analysis, and evaluation of sound recordings that may ultimately be presented as admissible evidence in a court of law or some other official venue.

Audio forensic evidence may come from a criminal investigation by law enforcement or as part of an official inquiry into an accident, fraud, accusation of slander, or some other civil incident.



Typical investigations involve one or more of three primary concerns:

***Authenticity,
enhancement,
interpreting .***

Types of Audio Forensic Investigations

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Authenticity:

The primary step for the analysis of an audio recording is to establish the authenticity of the recording. The forensic examiner verifies if any alteration such as addition, substitution or deletion has been made to the recording.

Enhancement:

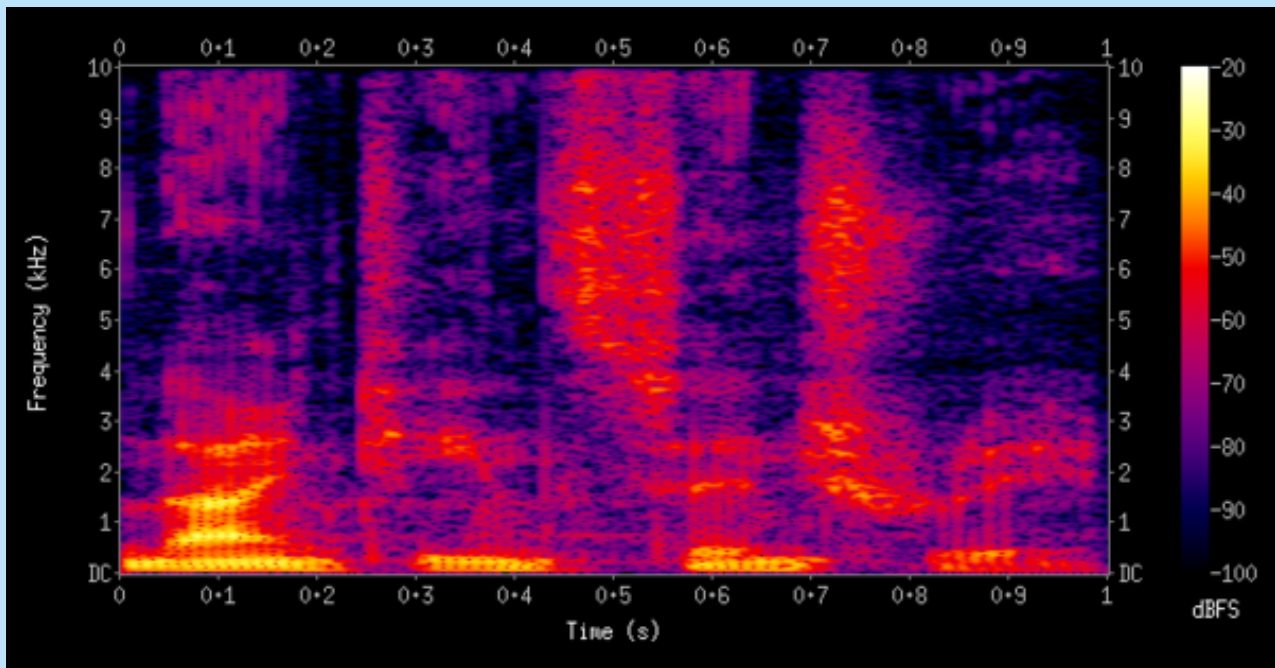
The quality of the audio evidence is not always good. Many times it is really difficult to recognize the speech in the recording due to the background noise and to resolve this issue, to understand what is being said, the enhancement of the audio recording is done.

(Frequency Equalization,. Compression)

Interpretation:

After authentication and enhancement, the evaluation of the audio recording is done in order to understand and interpret the relevance of the audio to the investigation. It includes recognition of the speech (speech recognition), recognition of the speaker (speaker identification), and interpretation of background noise that can indicate the environment in which the audio was recorded.

Spectrogram



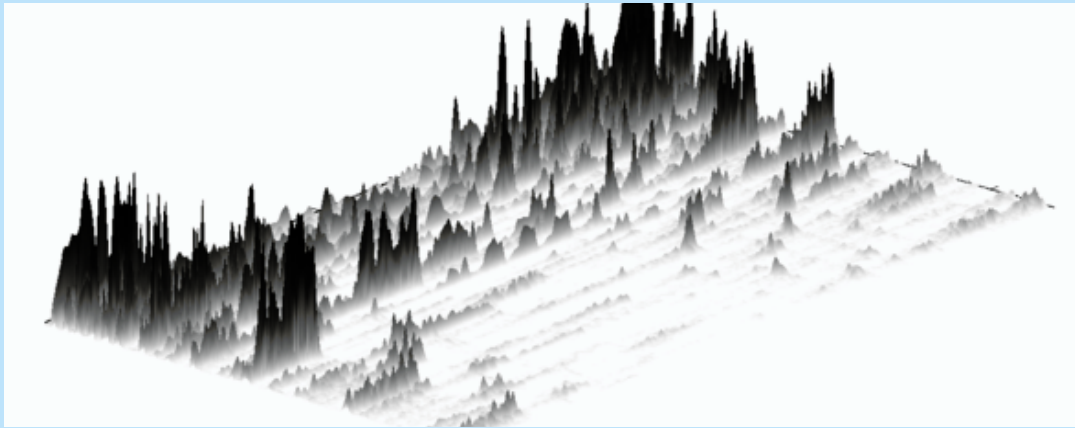
A spectrogram is a visual representation of the spectrum of frequencies of a signal as it varies with time. When applied to an audio signal, spectrograms are sometimes called sonographs, voiceprints, or voicegrams. When the data are represented in a 3D plot they may be called waterfalls

Spectrograms are used extensively in the fields of music, linguistics, sonar, radar, speech processing, seismology, and others. Spectrograms of audio can be used to identify spoken words phonetically, and to analyse the various calls of animals.

A spectrogram can be generated by an optical spectrometer, a bank of band-pass filters, by Fourier transform or by a wavelet transform (in which case it is also known as a scaleogram or scalogram).

Scaleograms from the DWT and CWT for an audio sample

A spectrogram is usually depicted as a heat map, i.e., as an image with the intensity shown by varying the colour or brightness.



***3D Surface Spectrogram of a part
from a Music piece.***

Audio Forensic Investigations

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***We have provided expert digital forensic
evidence and court room witness
testimony reports on speaker identity,
and audio authentication for a range of
high-profile cases in the following areas:***

Fraud

Counter-terrorism

Murder

Kidnap

Blackmail

Sexual offences

Drug offences

Civil litigation

Trading standards

Professional conduct investigations

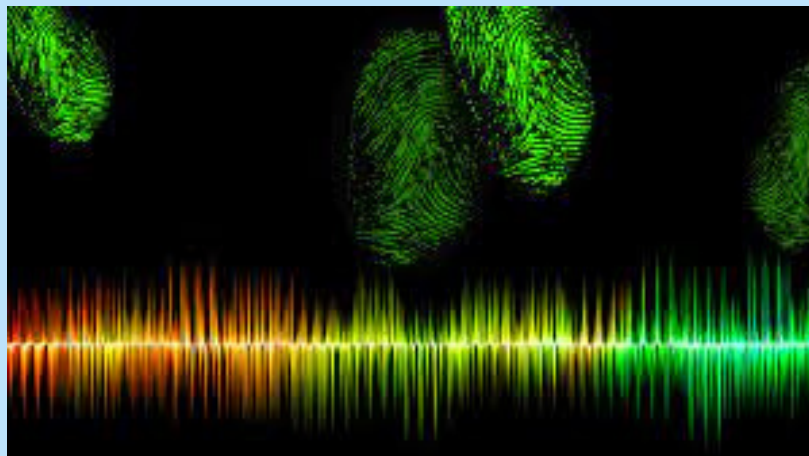
Employee investigations

Commercial disputes

Whistleblowing investigations

Bribery cases

Corruption



Audio Analysis Expert➤

Audio forensics is the close examination of audio files for legal purposes. Audio forensic experts examine the recording in a forensics lab to determine its authenticity. The examination typically involves checking integrity, improving speech clarity, transcribing dialogue, comparing voices, and reconstructing event timelines.

Analysis of Audio Evidence➤

One of the primary tasks of forensic digital investigators to assist the crime scene investigators in order to find the conclusive proof via a number of scientific tools and equipments.

After following the standardized procedure of crime scene investigation, at the time of evidence collection, the investigators must thoroughly search the suspected area and recover the evidence carefully. Such digital evidences must be protected from physical harm, environment, and heat.

Once the evidence is collected in a safe and secure manner, the proper documentation of evidence must be done in the form of notes or photography/videography. The documentation must include in which condition the evidence was found from the crime scene along with the name of the evidence collector, date, and time of evidence collection. All examination protocols are carefully examined and constructed for enhancement techniques that must be employed to the recovered evidence.

Forensic Speech Analysis Expertise

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Forensic speech analysis

Forensic voice analysis

Voice comparison

Voice identification

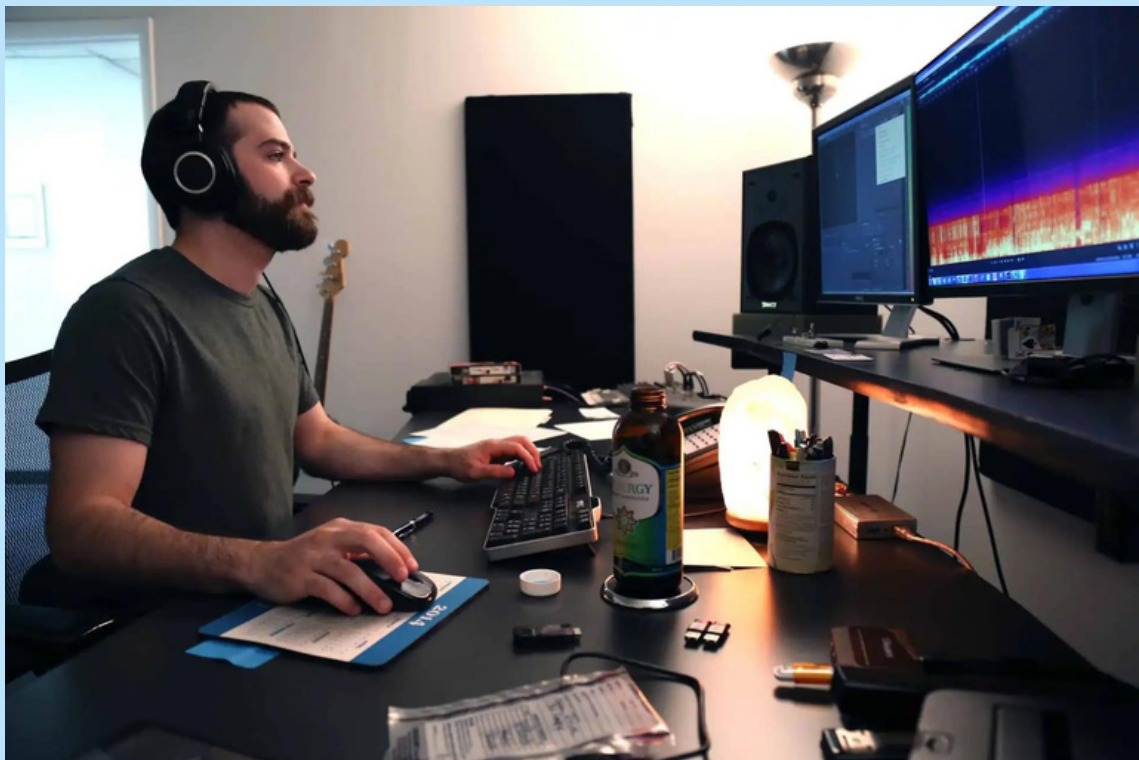
Speaker identification

transcription enhancement

Audio enhancement

Audio verification

Audio authentication



Advanced audio analysis

Recent advances in audio forensics techniques include voice biometrics and electrical network frequency analysis.

Voice biometrics:

Voice biometrics is the science of using a person's voice as a uniquely identifying biological characteristic in order to authenticate them. ... Voice biometrics also improves the customer experience by removing frustration associated cumbersome login processes and lost and stolen credentials

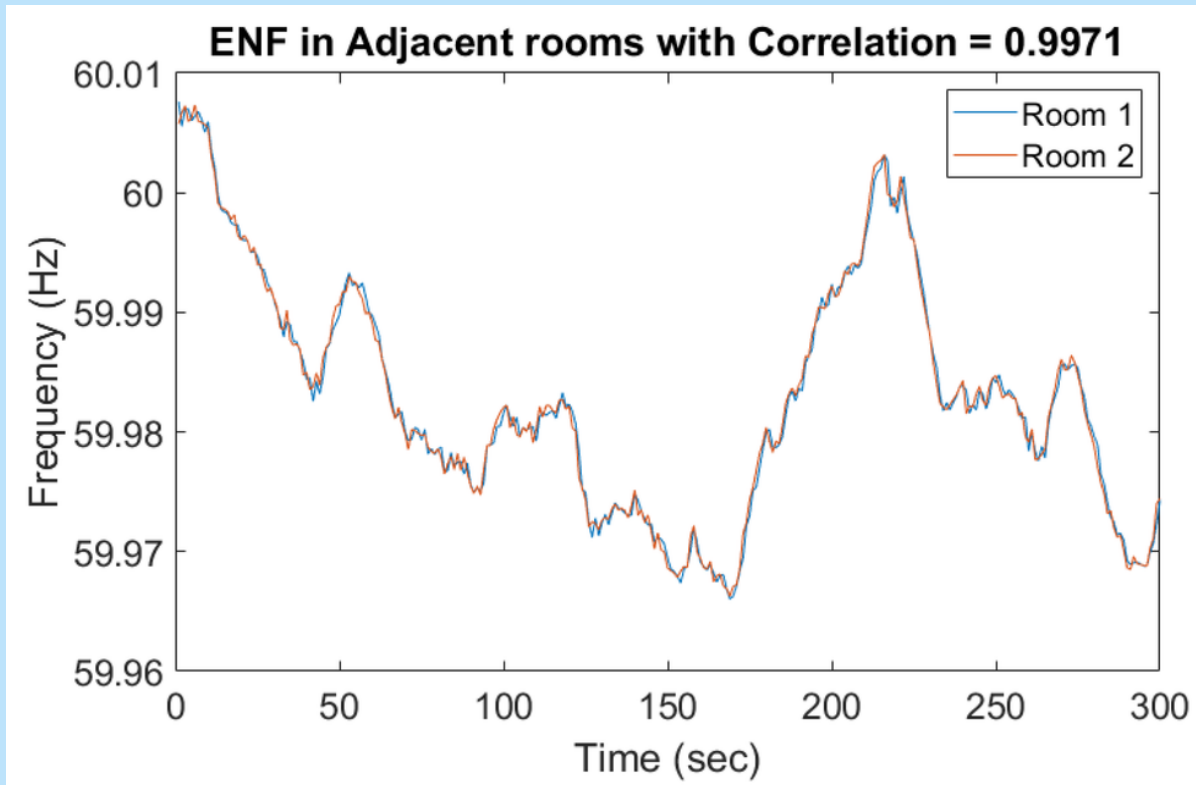
Voice biometrics



Electrical network frequency (ENF) :

Electrical network frequency (ENF) analysis is an audio forensics technique for validating audio recordings by comparing frequency changes in background mains hum in the recording with long-term high-precision historical records of mains frequency changes from a database.

Electrical network frequency (ENF) :



Audio analysis Report

The forensic audio-video report must include the following details:

- 1. Results which were obtained from the analysis of audio-video files***
- 2. Waveform charts of audio recordings and comparison waveform with formants.***
- 3. Identification of the format and type of recording***
- 4. Type of processing which was used to analyze.***
- 5. Date and Time of analysis of audio files.***
- 6. Description of the evidence in the type of circumstances and conditions in which it was collected***
- 7. Description of the enhanced audio-video and type of software that was used.***
- 8. Qualifications of the Audio Video Analyst.***
- 9. Authorized signatory with name and stamp.***

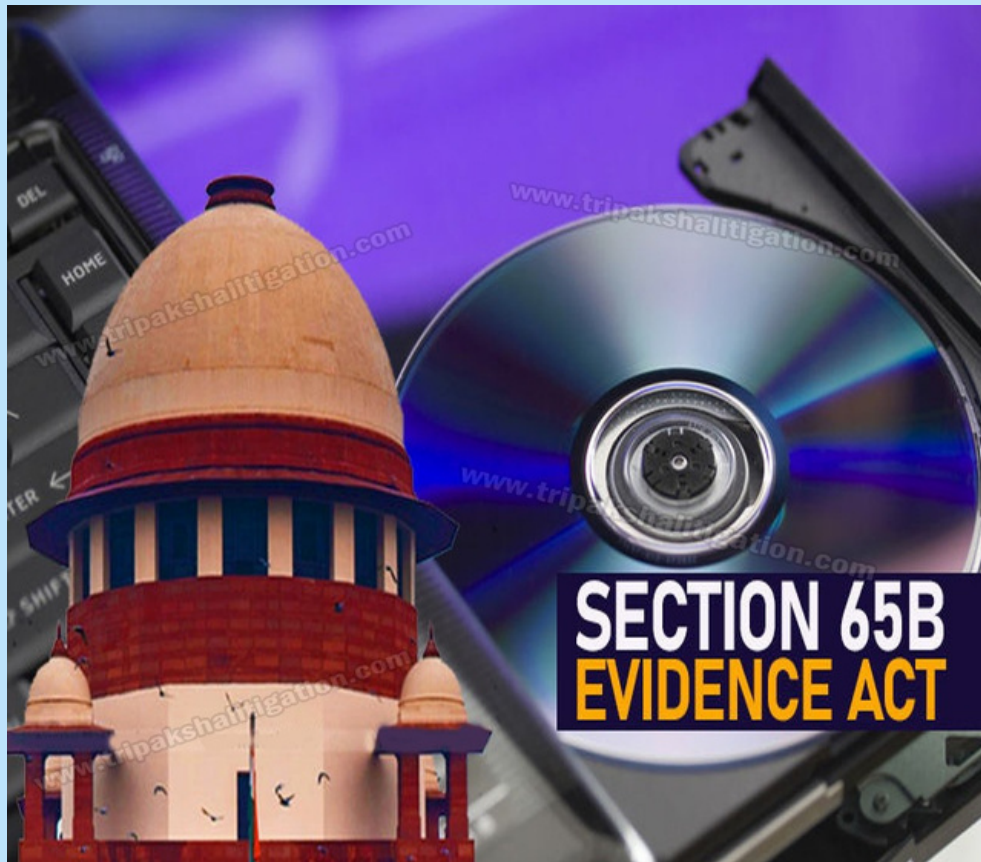


Law enforcement of India

According to Section 3 of the Indian Evidence Act, 1872, electronic records produced for the inspection of a court are considered as documentary evidence. Electronic records are admissible as per Section 65A and 65B of the same act. Due to these provisions, call recordings are admissible in a court of law.

In conclusion, phone audio recording is admissible as evidence in Courts under section 65 B of the Indian Evidence Act, 1872.

IEA sec 3 1872 65 B



In conclusion, phone audio recording is admissible as evidence in Courts under section 65 B of the Indian Evidence Act, 1872.



Role of audio Analysis

The primary aspects of audio forensics are establishing the authenticity of audio evidence, performing enhancement of audio recordings to improve speech intelligibility and the audibility of low-level sounds, and interpreting and documenting sonic evidence, such as identifying talkers, transcribing dialog, and reconstructing crime or accident scenes and timelines.



Scope of audio Analysis

- 1. Professors***
- 2. Associate and Assistant professors***
- 3. Researchers***
- 4. Lab instructors***
- 5. Trainers***
- 6. Legal advisors etc***

Typical job responsibilities

- Designing and teaching forensic science specific courses: both practical and theory based***
- Conducting and Overseeing forensic science based Research***
- Advising***
- Other administrative responsibilities***

Who they work for

- Community Colleges***
- Colleges and Universities***
- Federal, State and Local Law Enforcement***
- Academies***

How do I become a forensic audio analyst?

Both forensic expert and analyst positions require a master's degree in recording arts. Your degree program should also focus heavily on forensic science. Skills in audio enhancement, forensic litigation, and courtroom evidence preparation are essential to your success in these roles.



Software for Audio Analysis

WavePad Master

Sonic Visualizer

Praat

Gold Wave

Audacity

Speech Analyzer

**Frequently
asked
questions in
Google**

1 What is forensic audio analysis?

Forensic Speech Analysis involves the detailed analysis of audio, from sources such as a phone call, voicemail message or covertly recorded conversation. This is With the aim of producing a reliable phonetic profile of a speaker to determine the identity of said person.

2 Which of the following is an audio forensic tool?

A spectrogram shows both the frequency content of a recording and the level of those frequencies over time. It may be the most helpful tool to an audio forensic expert because it visually presents everything that is happening throughout the audio in one Window.

3 What is forensic audio specialist?

Audio forensic experts examine the recording in a forensics lab to confirm its authenticity and perform enhancement techniques to clarify its contents. The examination typically involves checking integrity, improving speech clarity, transcribing dialogue and reconstructing event timelines.

4 What is forensic audio evidence?

Audio forensics is the field of forensic science relating to the acquisition, analysis, and evaluation of sound recordings that may ultimately be presented as admissible evidence in a court of law or some other official venue. ... The field of audio forensics was largely established in 1973 during the Watergate scandal.

5 What does an audio forensics do?

The primary aspects of audio forensics are establishing the authenticity of audio evidence, performing enhancement of audio recordings to improve speech intelligibility and the audibility of low-level sounds, and interpreting and documenting sonic evidence, such as identifying talkers, transcribing dialog, and reconstructing crime or accident scenes and timelines.

6 What is forensic audio analysis?

Forensic Speech Analysis involves the detailed analysis of audio, from sources such as a phone call, voicemail message or covertly recorded conversation. This is with the aim of producing a reliable phonetic profile of a speaker to determine the identity of said person.

7 What is the role of a audio forensic expert?

They duties are to analyze audio evidence for authenticity, often refuting or pinpointing a crime. As an expert in the field, your responsibilities are to Work with analysts to improve the intelligibility of audio speech and the audibility of quiet sounds.



Reference

<https://cyfor.co.uk/digital-forensics/forensic-speech-analysis/>

<https://www.sifs.in/blog-details/digital-audio-forensics/48>

...THE END...

